

KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY

Deemed to be University, Bhubaneswar - 751024, Odisha

MID SEMESTER EXAMINATION — EXAMINATION QUESTION PAPER

Subject: Data Structure

Time: 1.5 Hours

Semester / Programme: 3rd Semester

Full Marks: 20

Instructions: Answer Question No. 1 (Compulsory) and any THREE from Question No. 2 to 5. Figures in brackets indicate marks.

SECTION - A (Compulsory)

1(a) How many different Binary Search Trees (BST) can be constructed with 4 distinct keys? [1]

$$C_n = \frac{1}{n+1} \binom{2n}{n} = \prod_{i=2}^n \frac{n+i}{i}$$

1(b) State the worst-case time complexity of searching an element in a Binary Search Tree (BST) containing n nodes, and specify the condition under which it occurs. [1]

$$T(n) = O(n)$$

1(c) Evaluate the postfix expression: 5 \; 3 \; + \; 8 \; 2 \; - \; \; * [1]

1(d) What is the primary advantage of a Circular Queue over a Linear Queue implemented using a static array? [1]

1(e) Define a sparse matrix and write its compressed triple form (Row, Column, Value) storage structure representation parameters for a matrix of size $M \times N$ with *non-zero* elements K . [1]

$$\text{Storage} = (K + 1) \times 3$$

SECTION - B (Answer any THREE)

Answer any THREE questions from Q.2 to Q.5. Each question carries 5 marks.

Q.2. (a) Write an efficient algorithm/C function to reverse a singly linked list iteratively. Explain its time and space complexity. [3]

(b) Explain how two sparse matrices can be added efficiently using compressed coordinate (triple) representation. Give the structural outline of the algorithm. [2]

Q.3. (a) Convert the following infix expression into its equivalent postfix expression using a stack showing the step-by-step trace: $Q = (A + B) * C - (D - E) / (F + G)$ [3]

(b) Write a recursive function in C/C++ to find the height (or maximum depth) of a binary tree. [2]
 $\text{Height}(T) = 1 + \max(\text{Height}(T_{\text{left}}), \text{Height}(T_{\text{right}}))$

Q.4. (a) Construct a Binary Search Tree (BST) by inserting the following sequence of elements one by one: 50, 30, 70, 20, 40, 60, 80. Then, write the post-order traversal sequence of the resulting BST. [3]

(b) Write a pseudo-code or C function to traverse a binary tree in ****in-order**** using an explicit user-defined stack (non-recursive). [2]

- Q.5.** (a) Construct a unique Binary Tree from the given traversal sequences: - Preorder Traversal : `A B D E C F` [3]
- In-order Traversal : `D B E A F C`
- (b) Write a complete C/C++ function to insert a new node at the beginning of a Doubly Linked List (DLL), [2]
updating both pointers correctly.
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